

Named Entity Recognition with spaCy

Using Hugging Face Transformers to do Named Entity Recognition

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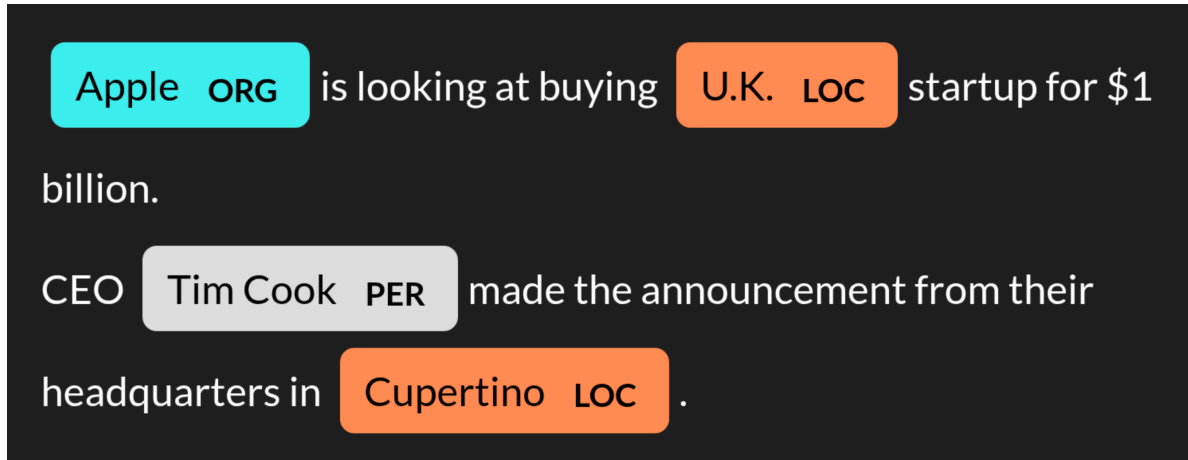


Figure 1: Named-entity recognition (NER) Example

Named-entity recognition (NER) is about identifying and classifying named entities in text into predefined categories such as **persons**, **organizations**, **locations**, etc. It extracts entities from unstructured text.

By identifying and categorizing real-world entities in text, NER bridges the gap between unstructured text and structured information that can drive decision-making and insights. Here's why it's so important:

1. Business Intelligence & Market Analysis

- **Competitive Intelligence:** Extract competitor names, products, and initiatives from news articles and reports
- **Market Trends:** Identify emerging players, technologies, and market shifts by analyzing mentions of organizations and products
- **Customer Insights:** Extract brand mentions, product names, and sentiment indicators from customer feedback

2. Information Extraction & Knowledge Management

- **Automated Data Population:** Populate databases and knowledge graphs with structured information extracted from unstructured text
- **Document Classification:** Enhance document categorization by understanding the entities mentioned within content
- **Content Recommendation:** Improve content recommendations by matching based on entities of interest to users

3. Search & Retrieval Enhancement

- **Semantic Search:** Enable entity-based search beyond simple keyword matching
- **Question Answering:** Power systems that can answer “who”, “where”, and “when” questions by extracting relevant entities
- **Document Summarization:** Identify key figures, organizations, and locations for inclusion in automated summaries

4. Healthcare & Biomedical Applications

- **Clinical Data Mining:** Extract patient information, conditions, treatments, and medications from clinical notes
- **Medical Literature Analysis:** Identify relationships between diseases, drugs, and genes in research papers
- **Clinical Trial Matching:** Match patients to appropriate clinical trials based on extracted medical entities

5. Finance & Risk Management

- **Regulatory Compliance:** Extract reporting entities and requirements from regulatory documents
- **Financial News Analysis:** Track mentions of companies, executives, and financial metrics in real-time
- **Risk Assessment:** Identify potentially problematic entities (people, organizations) for due diligence

6. Legal Document Analysis

- **Contract Analysis:** Extract parties, dates, monetary values, and conditions from legal documents
- **Case Law Research:** Identify precedents, judges, courts, and legal principles in legal opinions
- **Compliance Monitoring:** Extract regulated entities and requirements from regulatory documents

7. Security & Intelligence

- **Threat Detection:** Identify potentially suspicious individuals, organizations, or locations in communications
- **Network Analysis:** Build relationship graphs between entities mentioned in various documents
- **Predictive Analysis:** Use historical patterns of entity mentions to predict future events or trends

8. Benefits Across Applications

- **Automation:** Reduce manual data entry and information extraction
- **Consistency:** Apply uniform entity extraction rules across large document collections
- **Discovery:** Uncover hidden relationships between entities across disparate sources
- **Time Efficiency:** Process volumes of text at scale that would be impossible manually
- **Structured Data Creation:** Transform unstructured text into structured, queryable information

Example

```
from transformers import AutoTokenizer, AutoModelForTokenClassification, pipeline
import spacy
from spacy import displacy
```

Initialize the NER processor with a transformer model using the (dslim/bert-base-NER).

Alternative model names for NER

- "jean-baptiste/roberta-large-ner-english"
- "dbmdz/bert-large-cased-finetuned-conll03-english"
- "elastic/distilbert-base-cased-finetuned-conll03-english"

For domain-specific NER, look for models trained on data from your domain.

For example, "[emilyalsentzer/Bio_ClinicalBERT](#)" for medical text.

```
model_name="dslim/bert-base-NER"

tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForTokenClassification.from_pretrained(model_name)
ner_pipeline = pipeline(
    "ner",
    model=model,
    tokenizer=tokenizer,
    aggregation_strategy="simple")
```

Some weights of the model checkpoint at dslim/bert-base-NER were not used when initializing LstmForTokenClassification
- This IS expected if you are initializing BertForTokenClassification from the checkpoint of LstmForTokenClassification
- This IS NOT expected if you are initializing BertForTokenClassification from the checkpoint of LstmForTokenClassification
Device set to use cpu

Extract named entities from text

```
text = """Apple is looking at buying U.K. startup for $1 billion.
CEO Tim Cook made the announcement from their headquarters in Cupertino.
"""
entities_list = ner_pipeline(text)
entities_list
```

```
[{'entity_group': 'ORG',
  'score': np.float32(0.9989743),
  'word': 'Apple',
  'start': 0,
  'end': 5},
 {'entity_group': 'LOC',
  'score': np.float32(0.98070335),
  'word': 'U. K.',
  'start': 27,
  'end': 31},
 {'entity_group': 'PER',
  'score': np.float32(0.9997685),
  'word': 'Tim Cook',
  'start': 61,
  'end': 69},
```

```
{'entity_group': 'LOC',  
  'score': np.float32(0.9980777),  
  'word': 'Cupertino',  
  'start': 119,  
  'end': 128}]
```

Visualize entities in text using spaCy's displacy.

```
nlp = spacy.blank("en")  
# Create a spaCy Doc  
doc = nlp(text)  
  
# Build spaCy Span objects for each entity  
spans = [  
    doc.char_span(  
        start_idx = ent["start"],  
        end_idx = ent["end"],  
        label = ent["entity_group"],  
        alignment_mode = "contract")  
    for ent  
    in entities_list]  
  
# Set the entities in the Doc  
doc.ents = filter(lambda x:x != None, spans)  
  
# Display using displacy  
displacy.render(doc, style="ent", jupyter=True)
```

Please visit <https://toknow.ai/posts/named-entity-recognition/index.html> to view the output of this code cell.

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